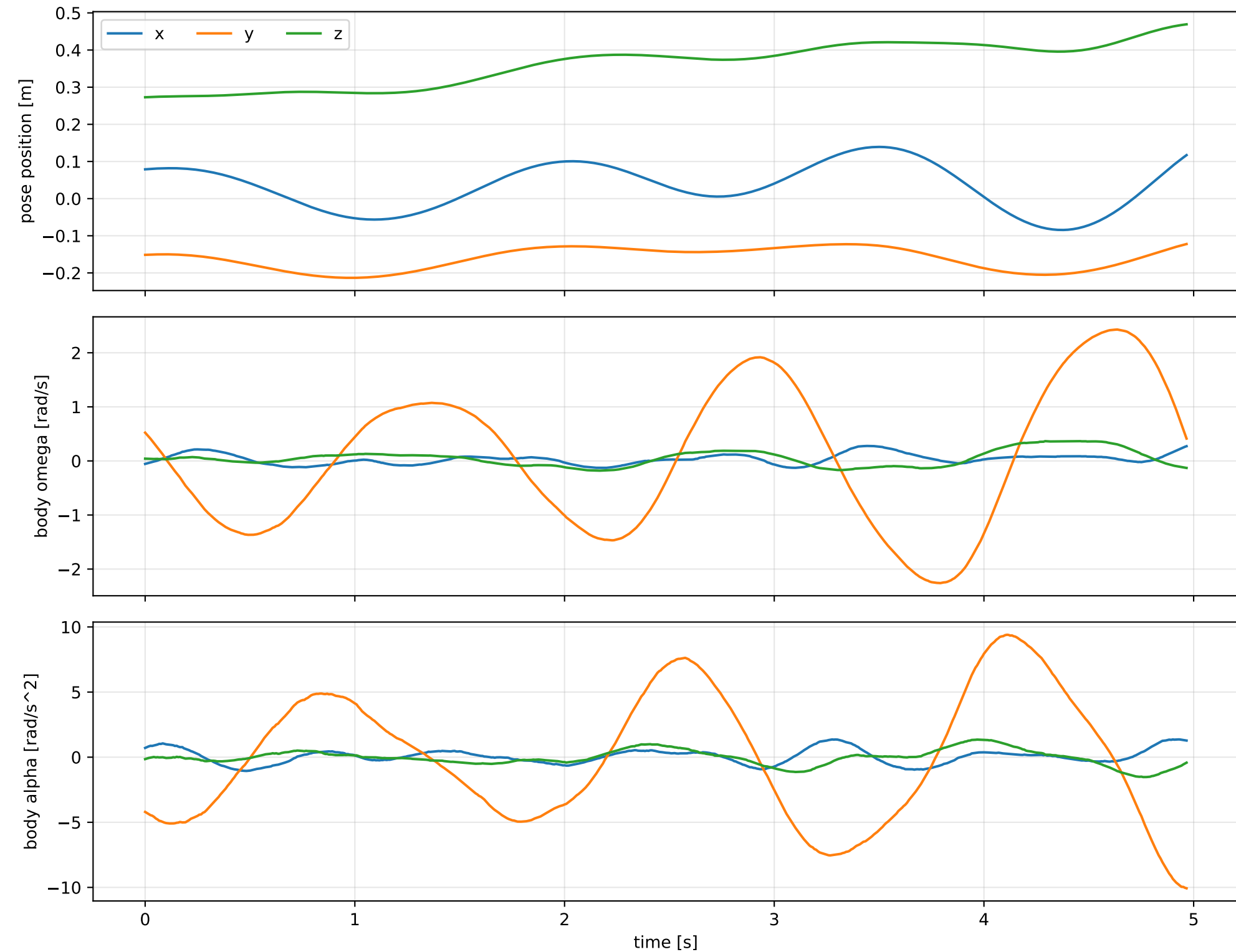
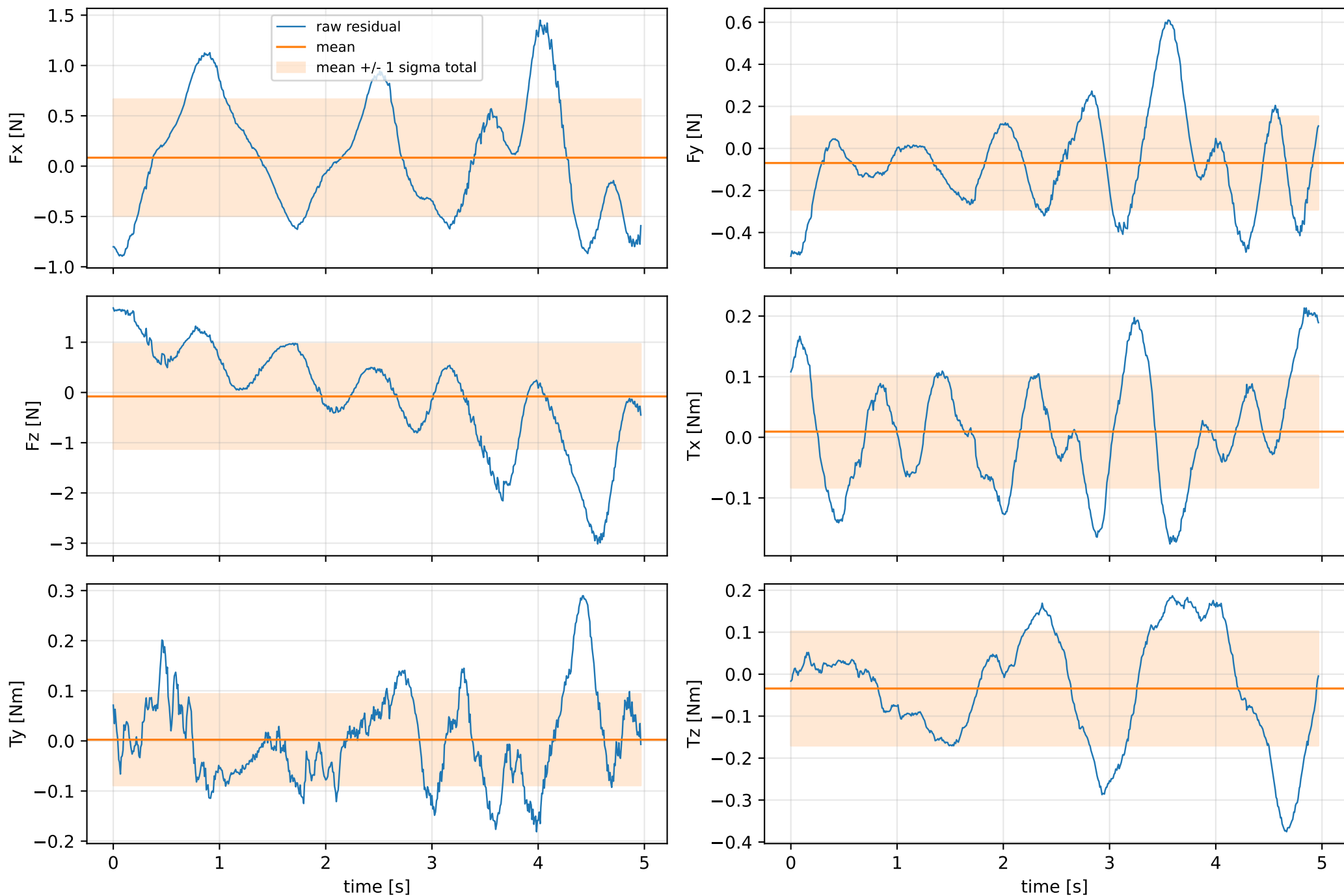


cog_all_nominal: SG trajectory and derived motion



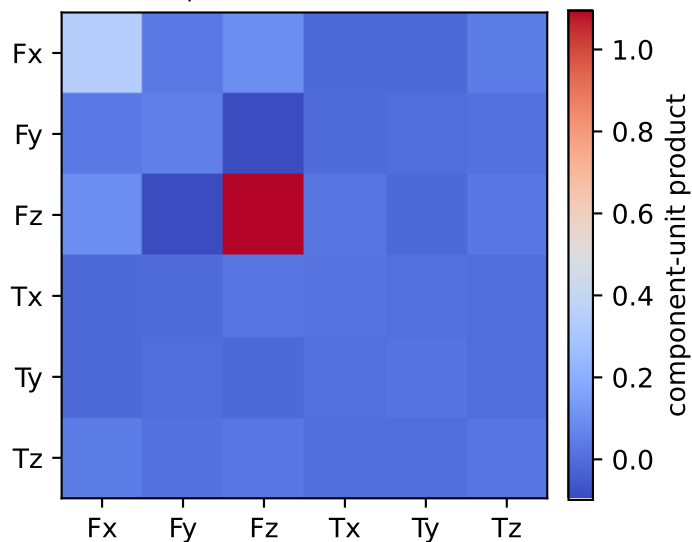
cog_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



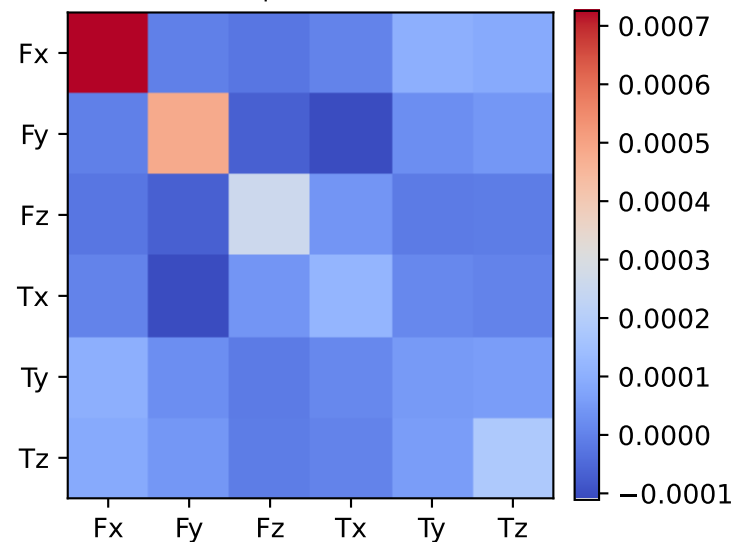
vector RMS about zero: force 1.2229 N, torque 0.19147 Nm; component std about mean: force [0.58 0.2223 1.046], torque [0.0926 0.0913 0.1362]

cog_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.004 0.007 0.013 0.039 0.337 1.114]

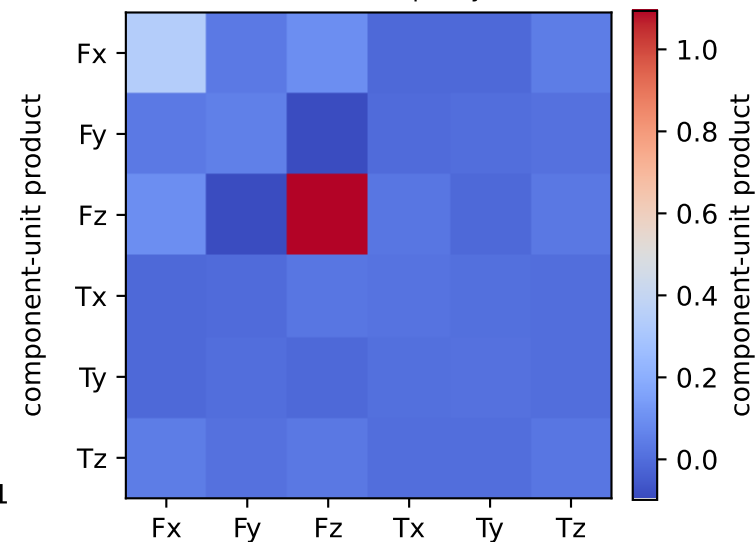
empirical total covariance



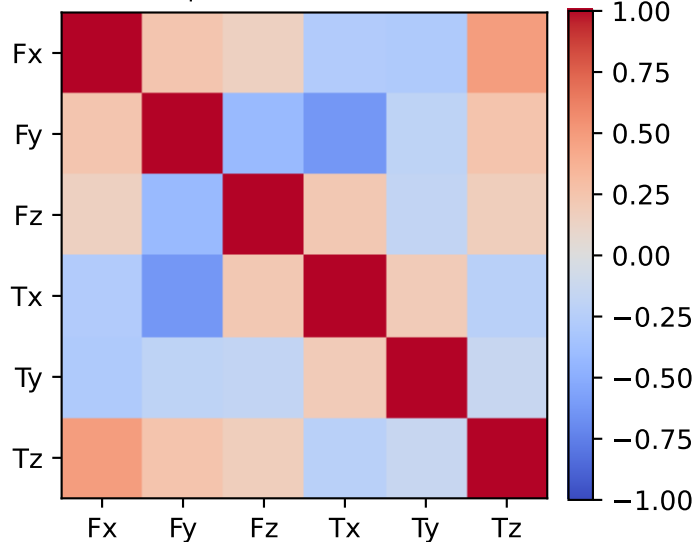
mean full-SG prediction covariance



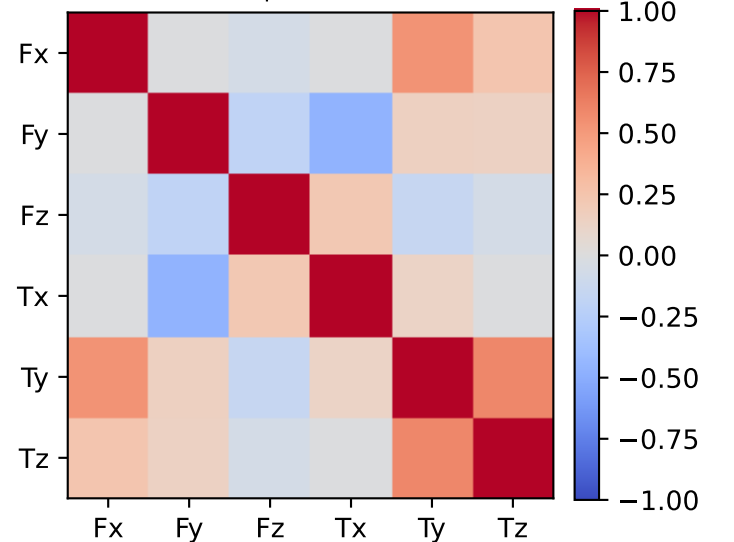
PSD excess model discrepancy covariance



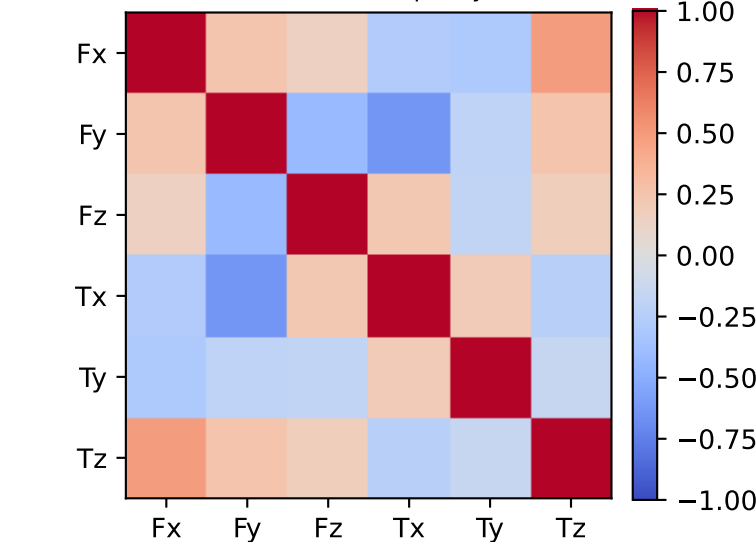
empirical total correlation



mean full-SG prediction correlation

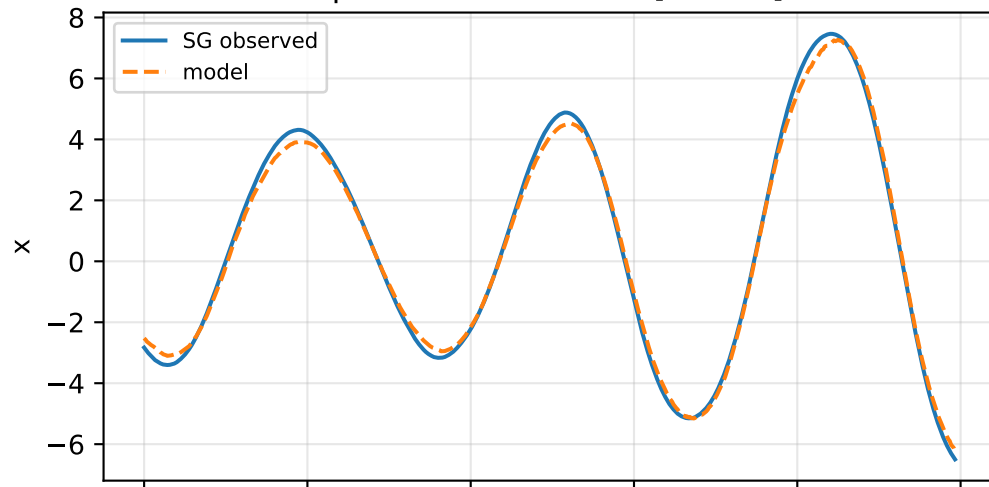


PSD excess model discrepancy correlation

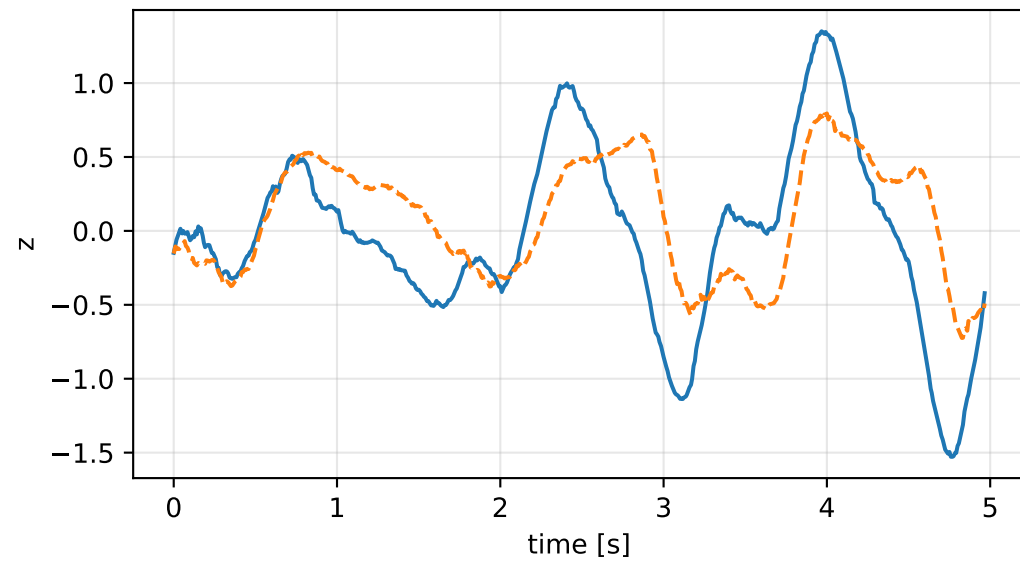
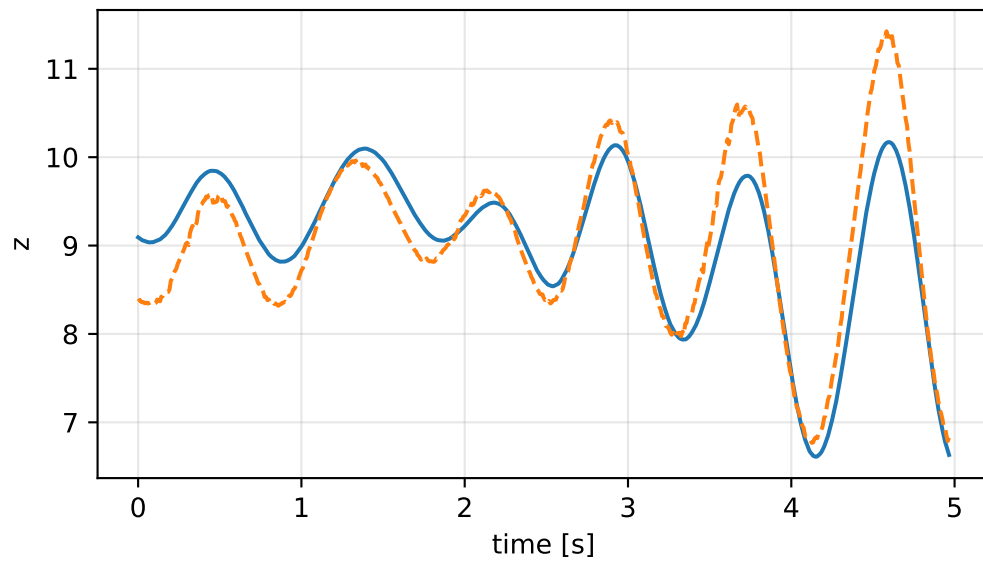
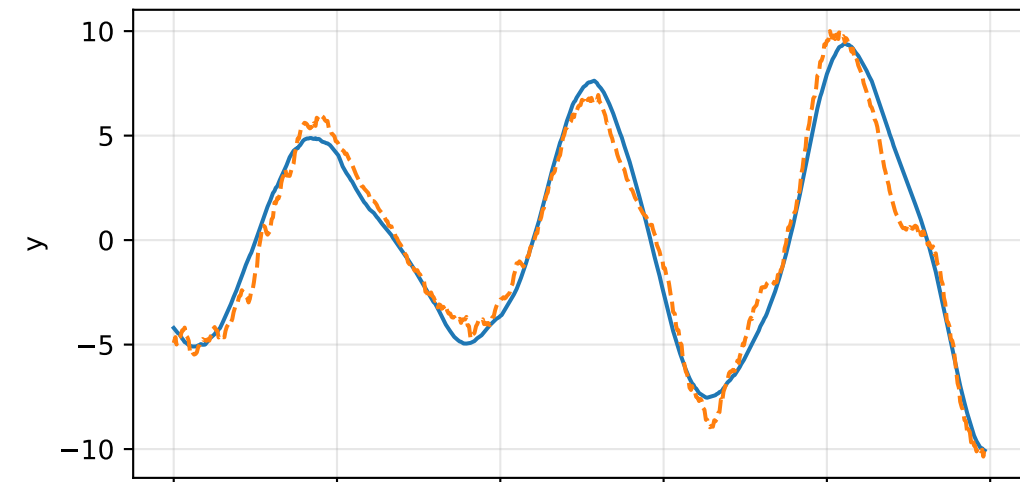
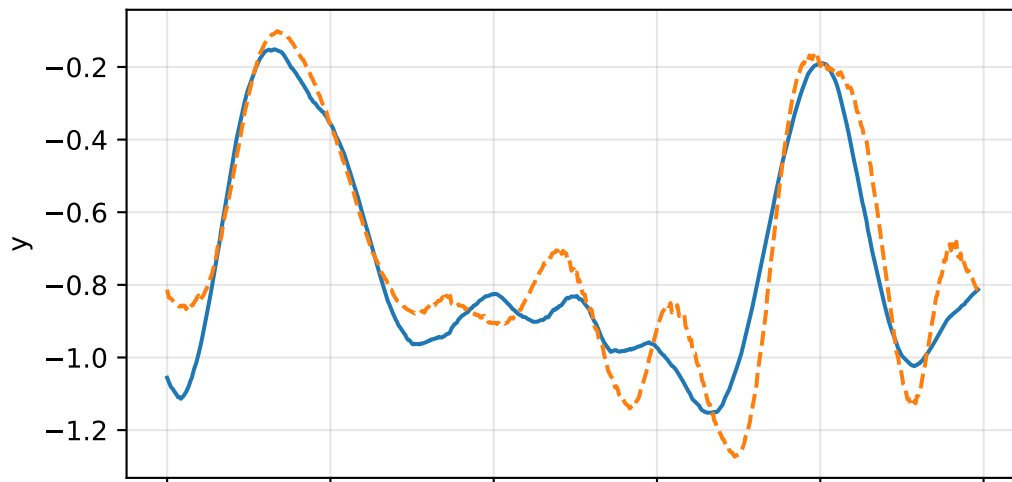
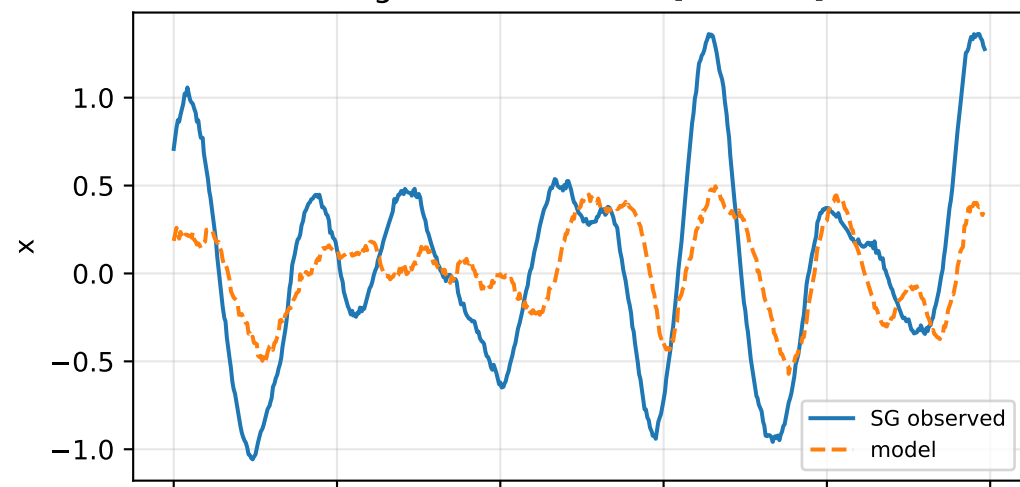


cog_all_nominal: acceleration residual objective

specific acceleration [m/s^2]

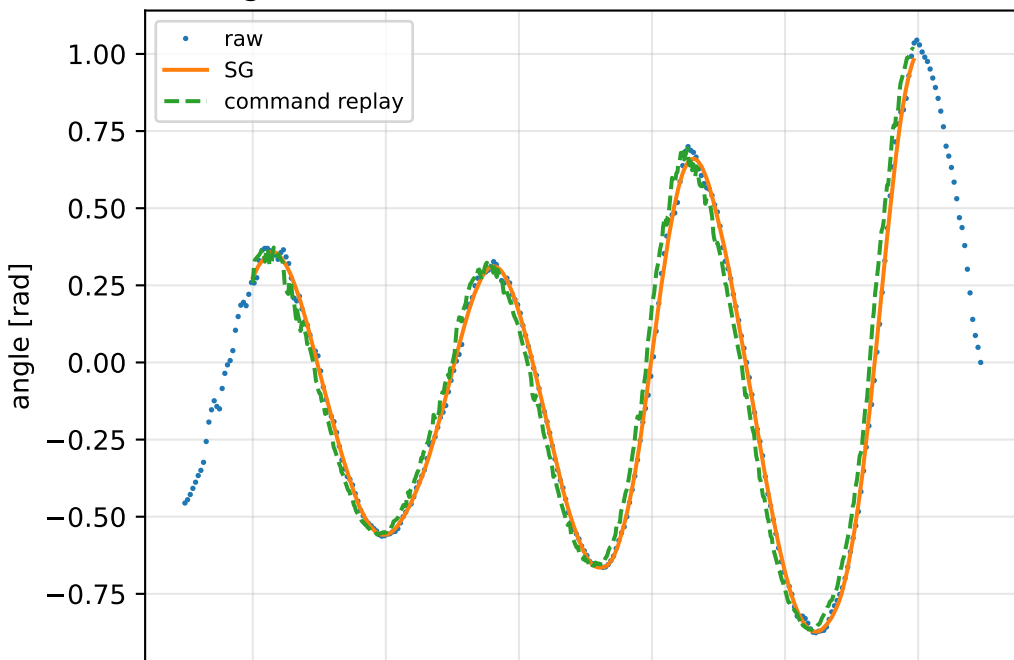


angular acceleration [rad/s^2]

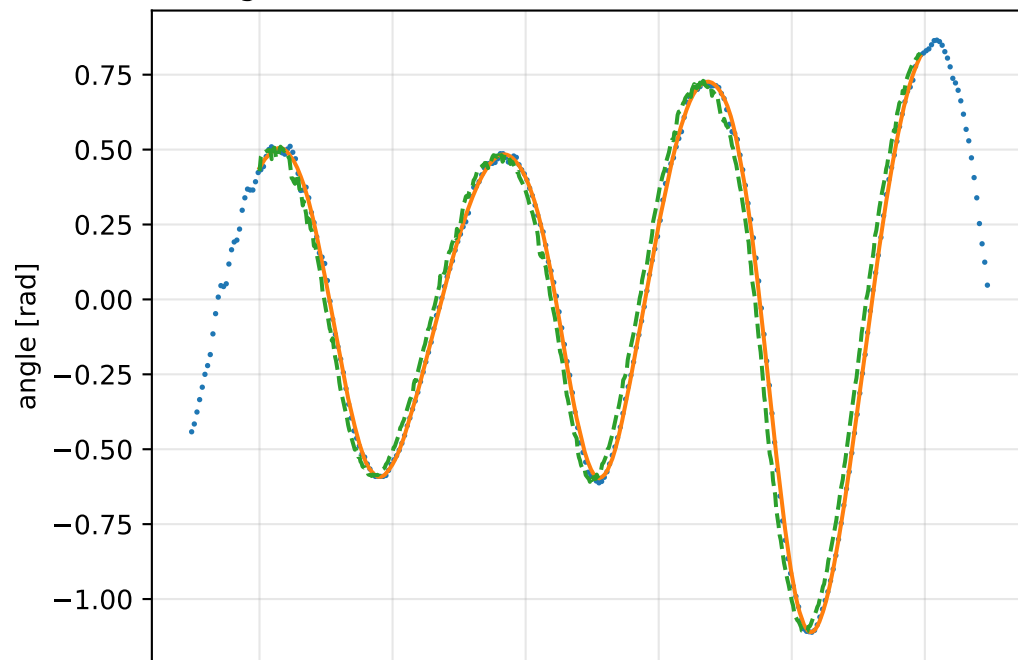


cog_all_nominal: gimbal measurement audit (objective=measured_sg)

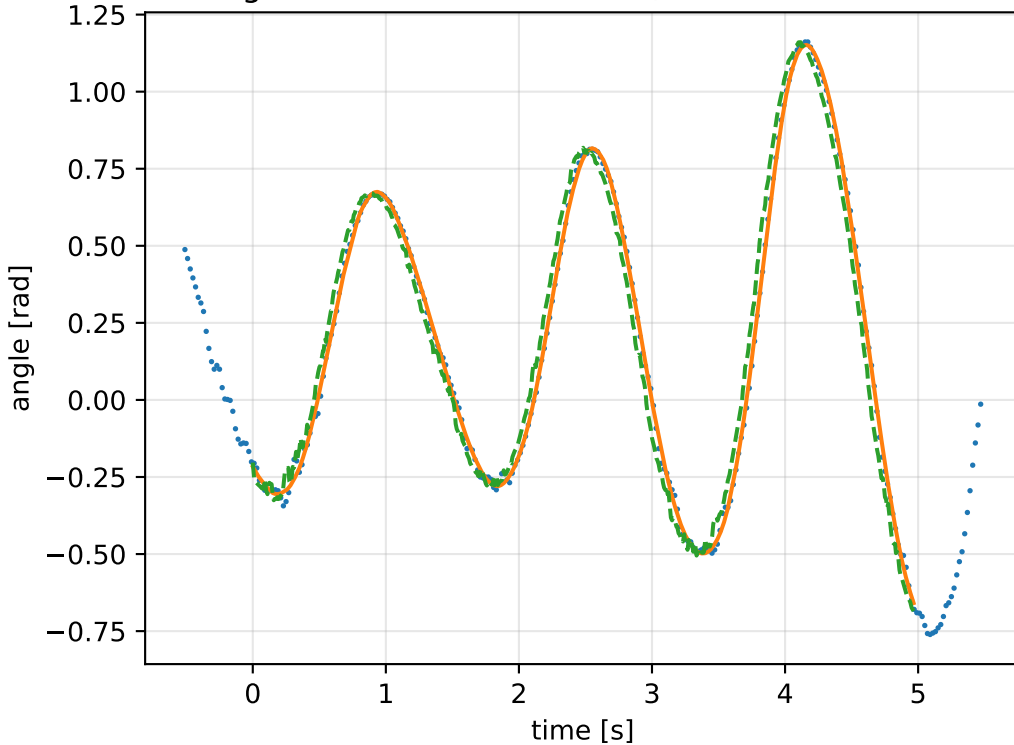
gimbal 1: RMSE=0.0759, max=0.2095 rad



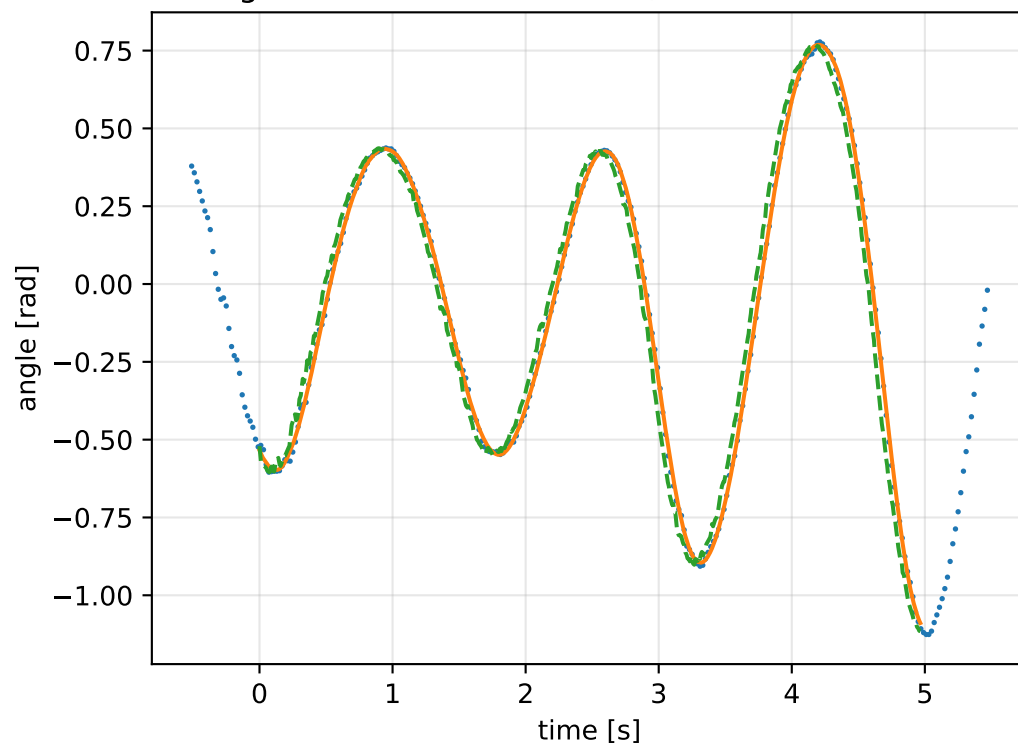
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

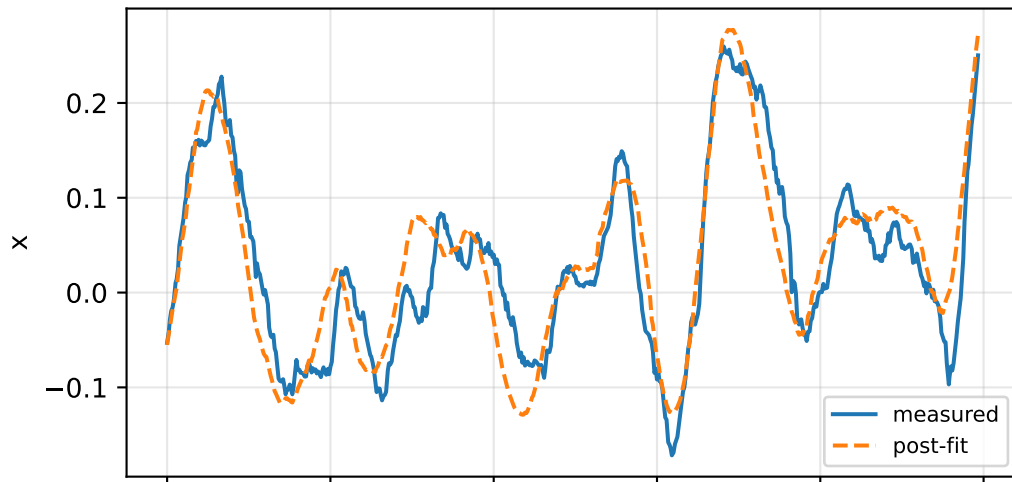


gimbal 4: RMSE=0.07131, max=0.1737 rad

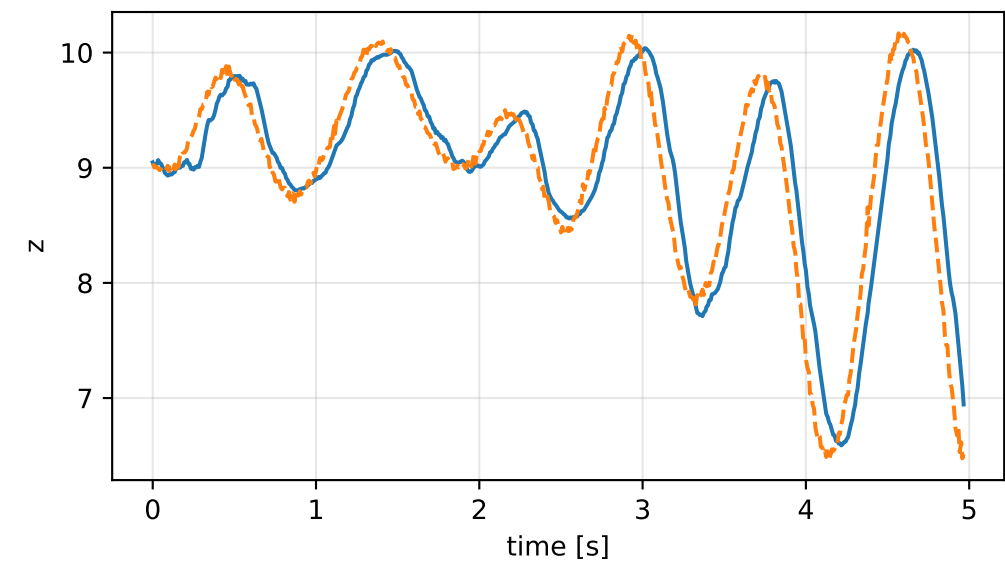
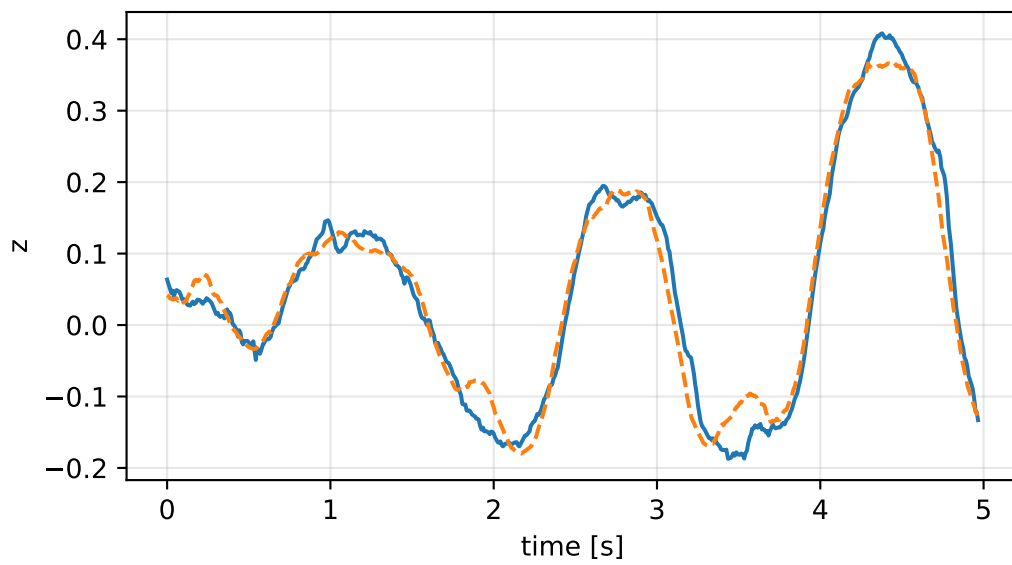
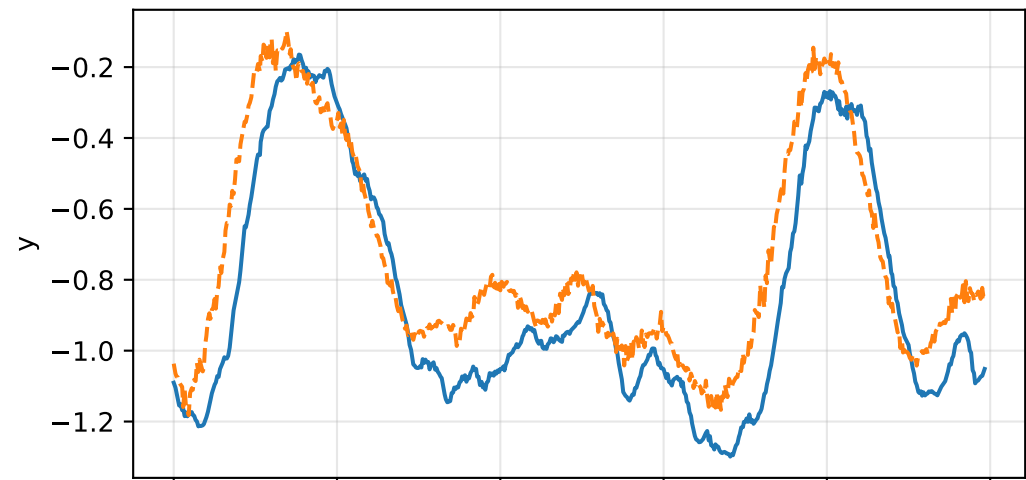
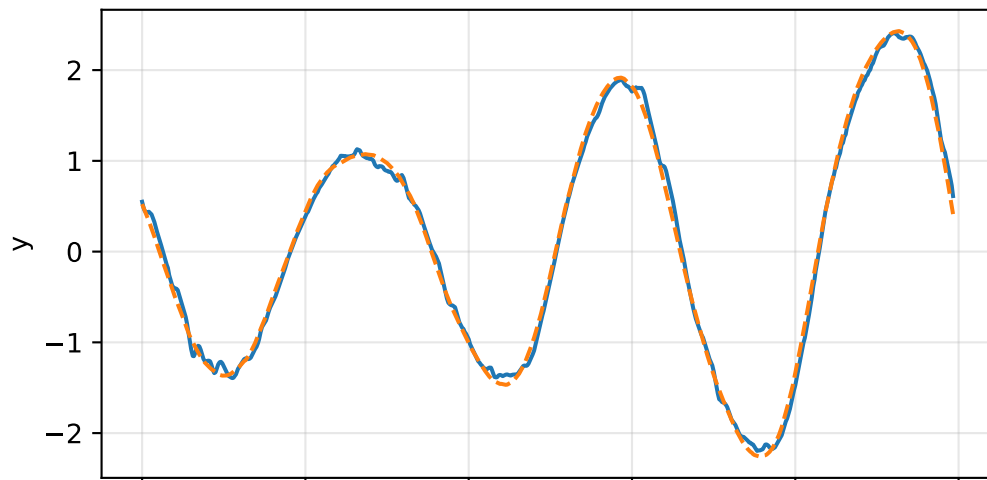
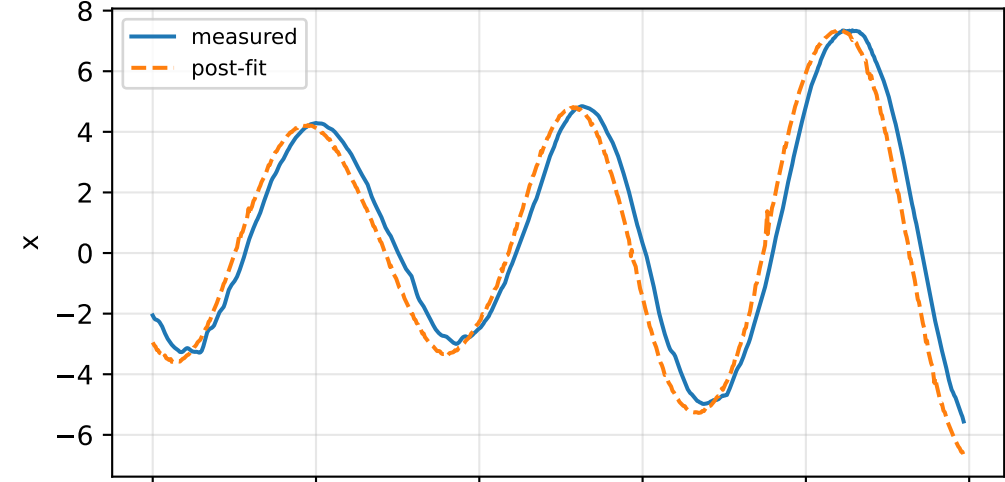


cog_all_nominal: IMU comparison (diagnostic only)

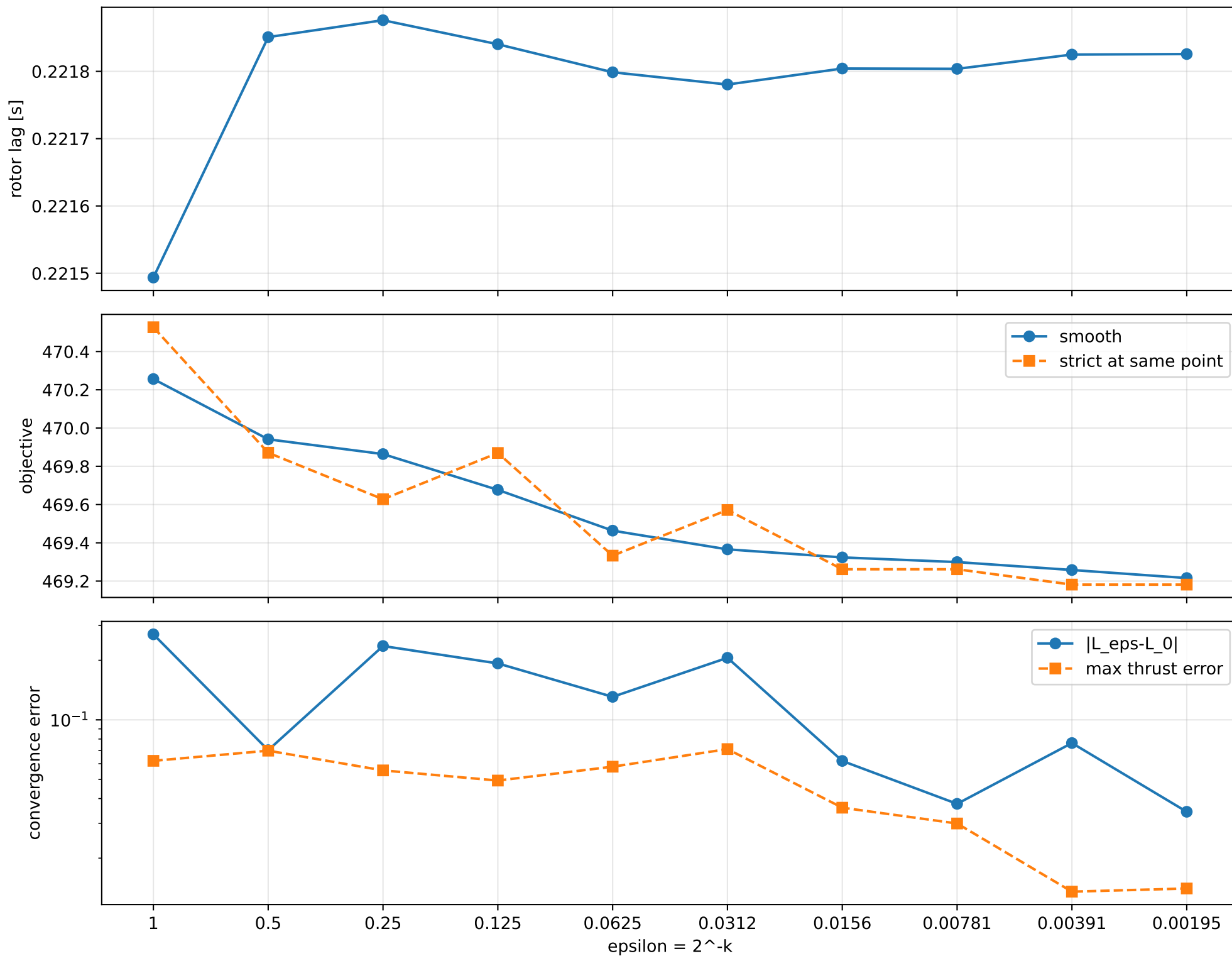
gyro [rad/s]



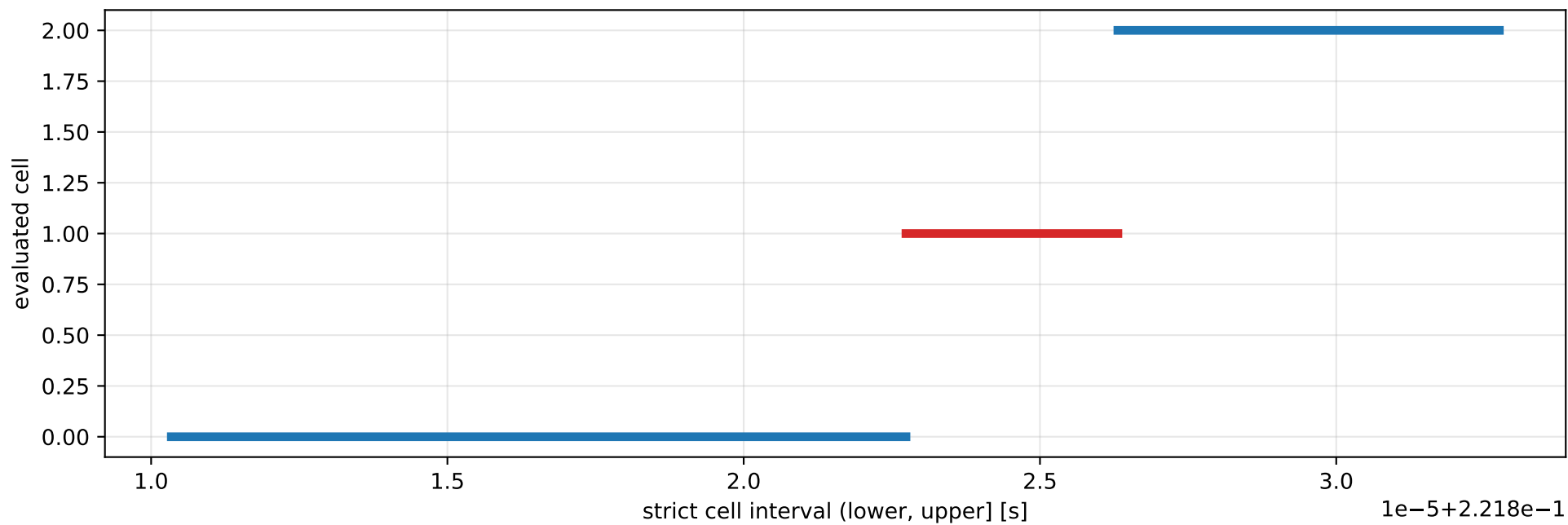
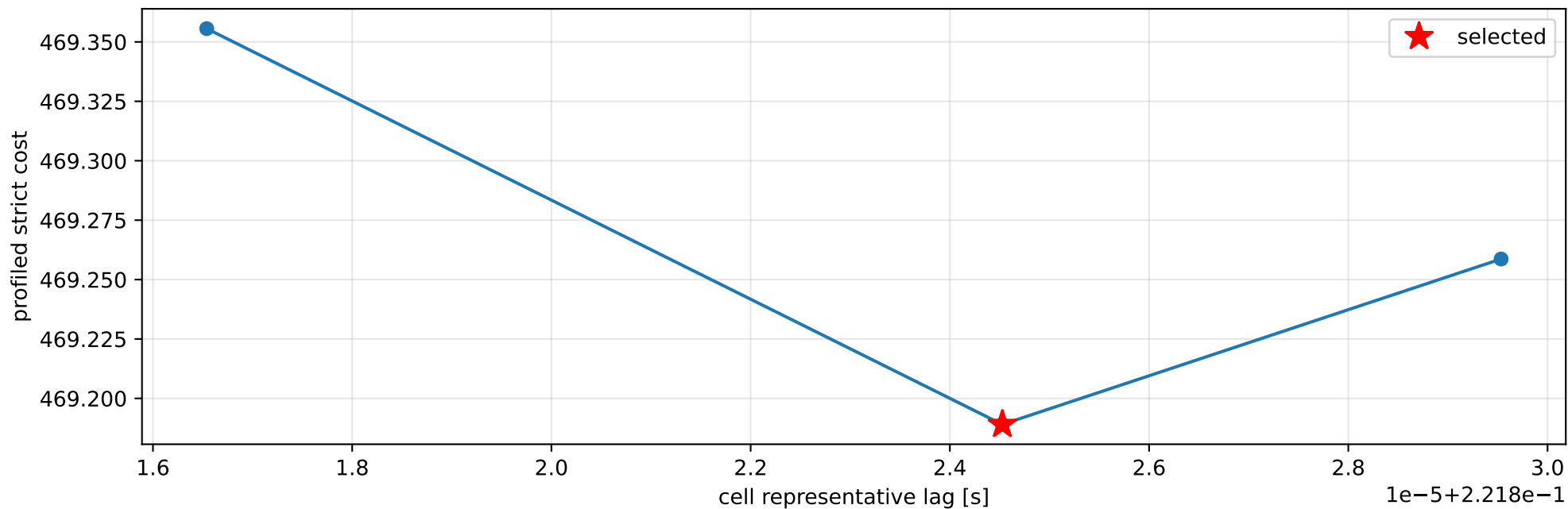
specific force [m/s^2]



cog_all_nominal: smooth-to-strict continuation



cog_all_nominal: exact strict-ZOH lag cells



selected (0.221822739, 0.221826315] s; final smooth 0.221825715 s; data boundary=False

cog_all_nominal: parameter estimate

Case: cog_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.84503628

inertia [kg m²]:

```
[[ 0.33209575  0.00452461 -0.0432542 ]
 [ 0.00452461  0.1596355  -0.00976846]
 [-0.0432542  -0.00976846  0.46539654]]
```

CoG body [m]: [-2.02460688e-03 -3.04869956e-05 9.50853128e-03]

force effectiveness: [1.36888898 1.47694895 1.34118231 1.3920427]

scale-free J/m [m²]:

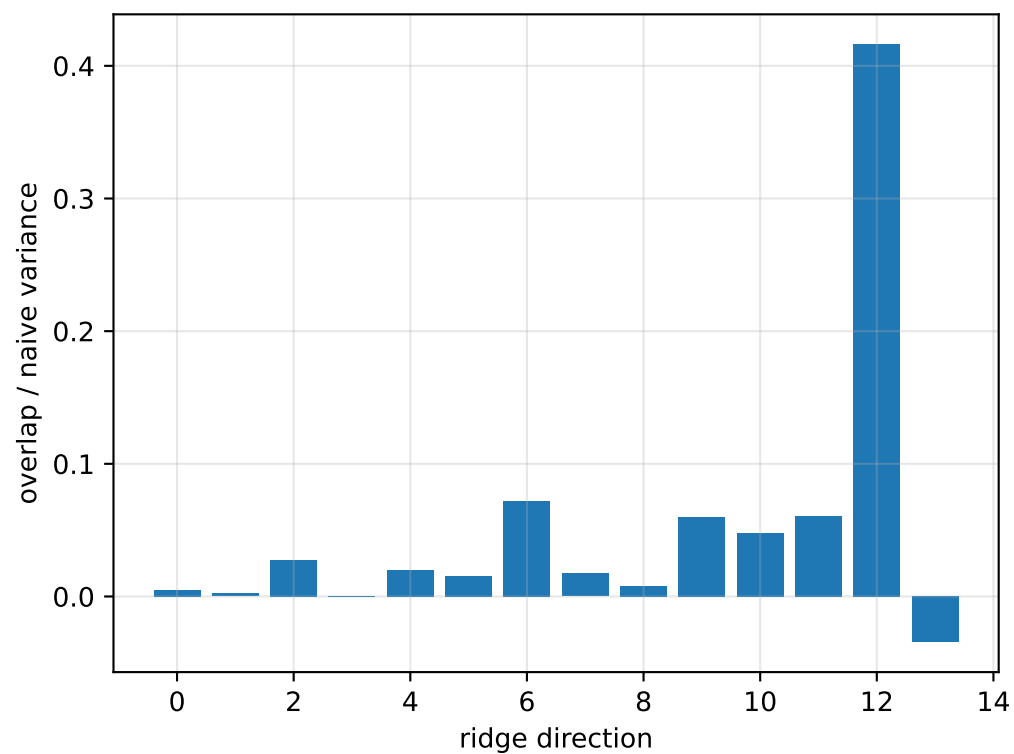
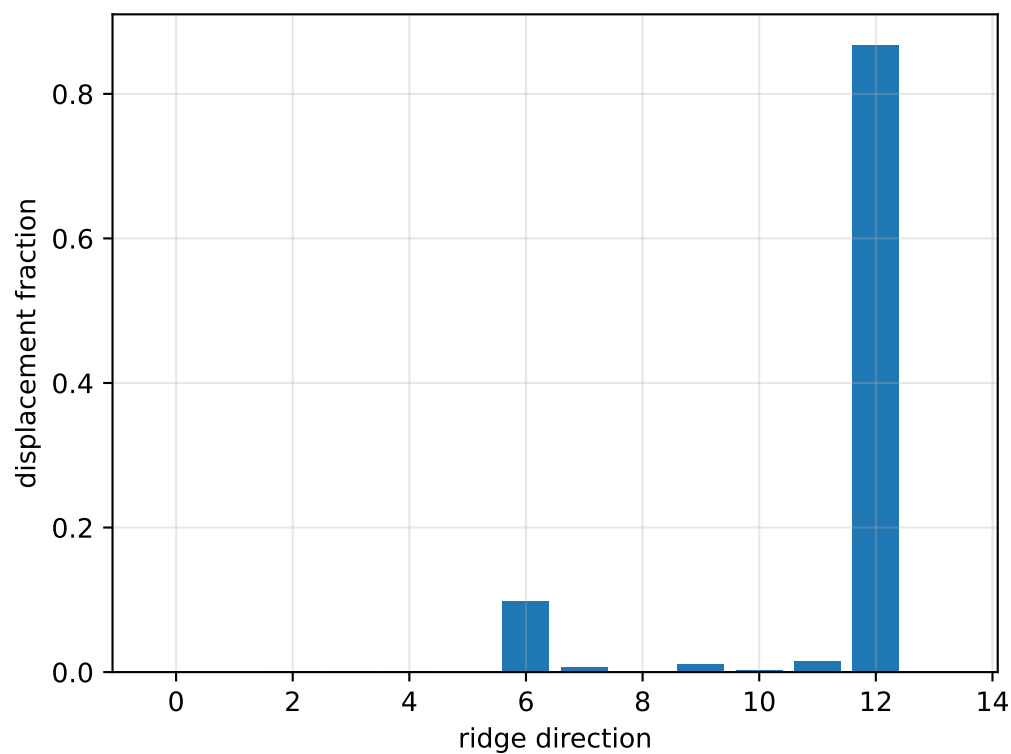
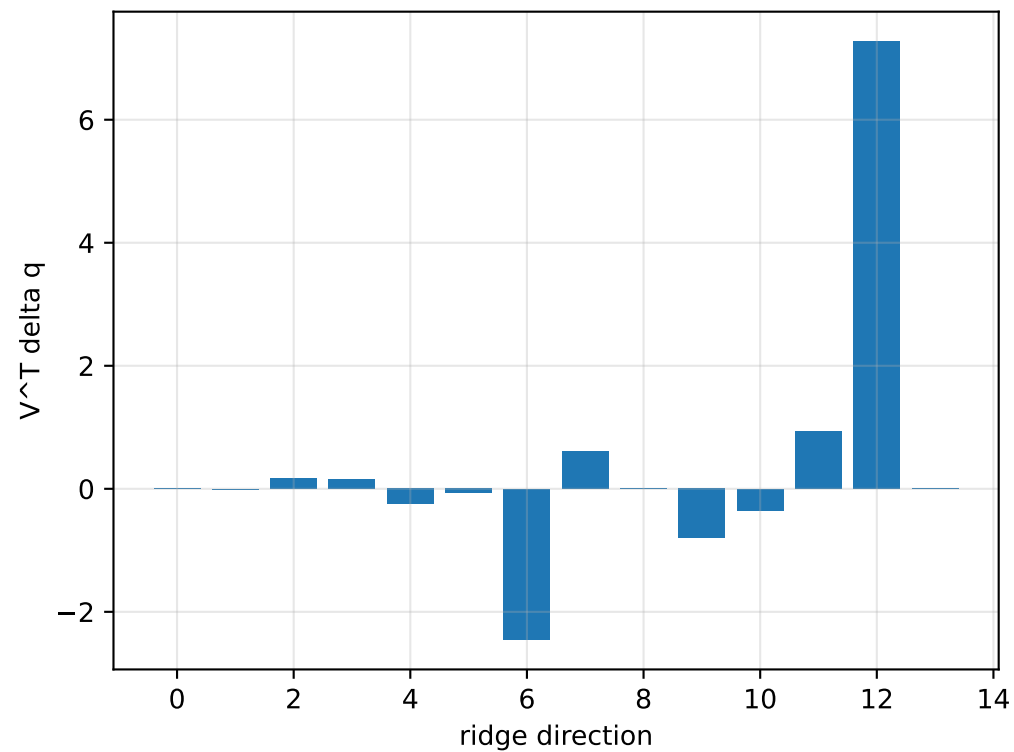
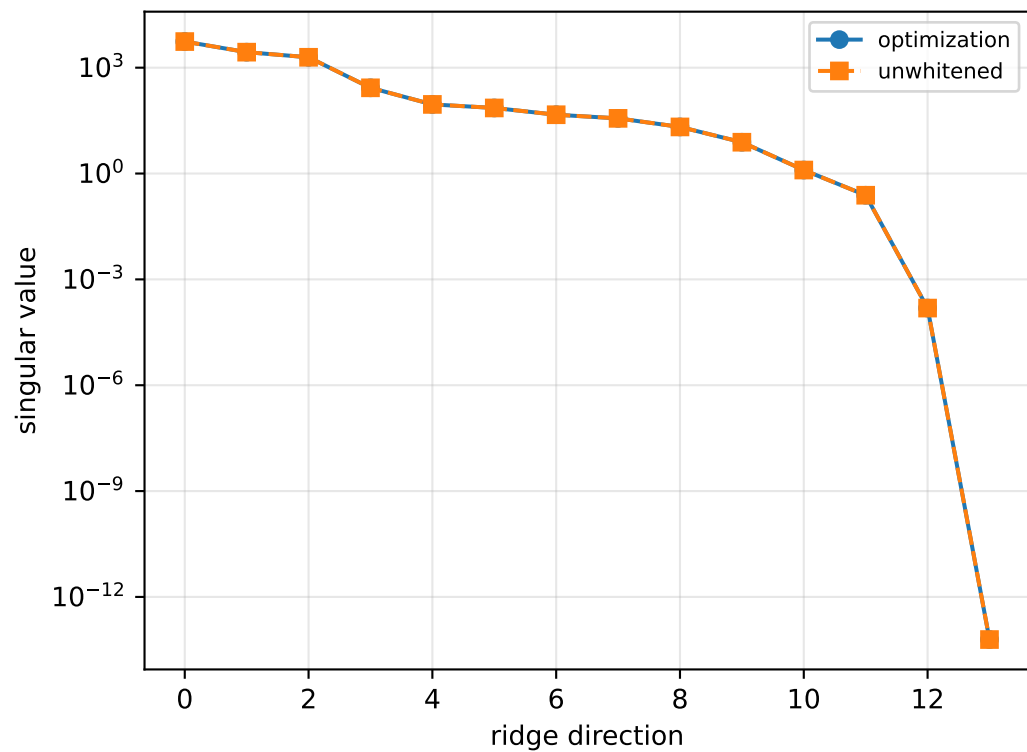
```
[[ 0.08636999  0.00117674 -0.01124936]
 [ 0.00117674  0.04151729 -0.00254054]
 [-0.01124936 -0.00254054  0.12103827]]
```

scale-free f/m: [0.35601458 0.38411834 0.34880875 0.3620363]

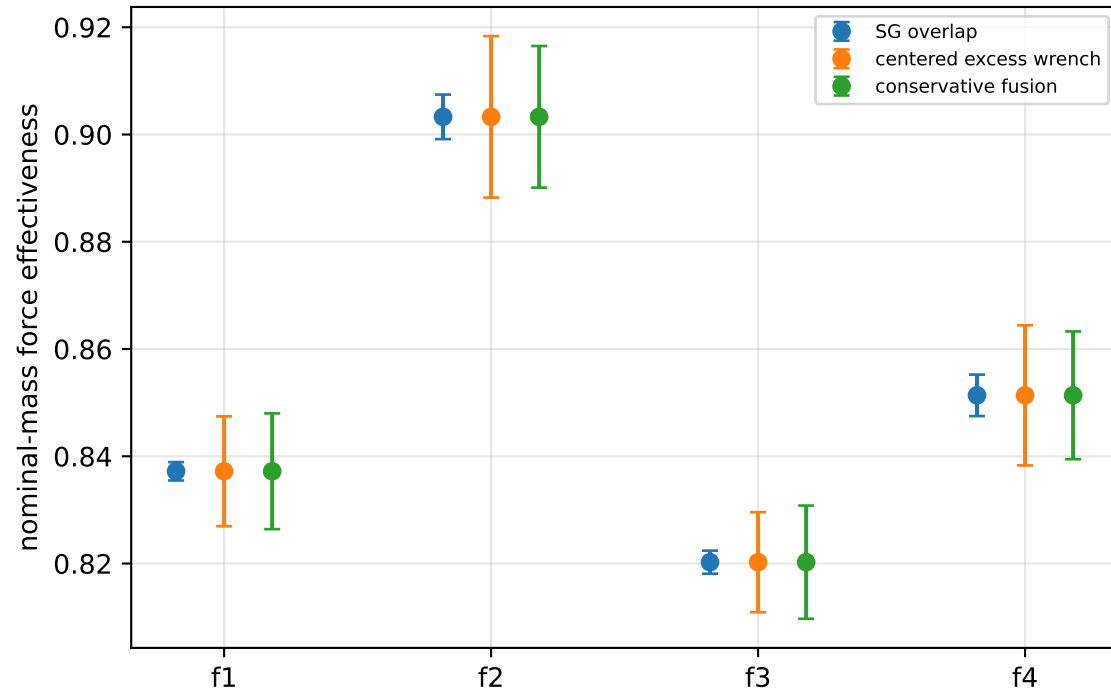
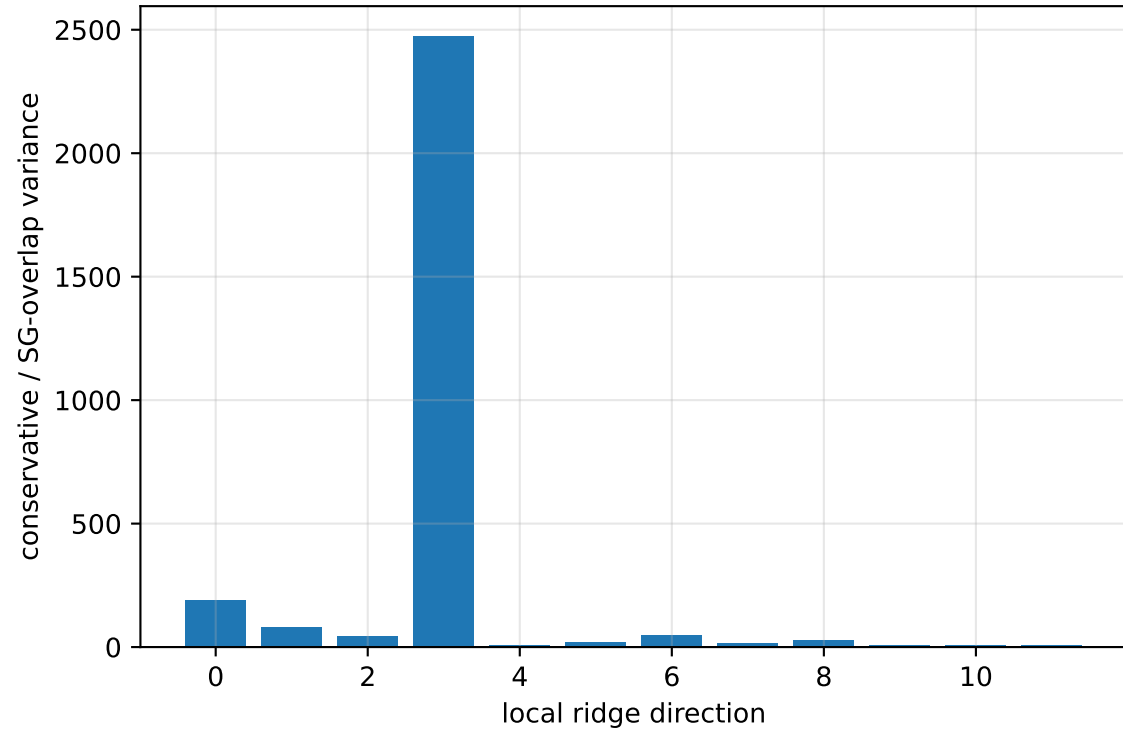
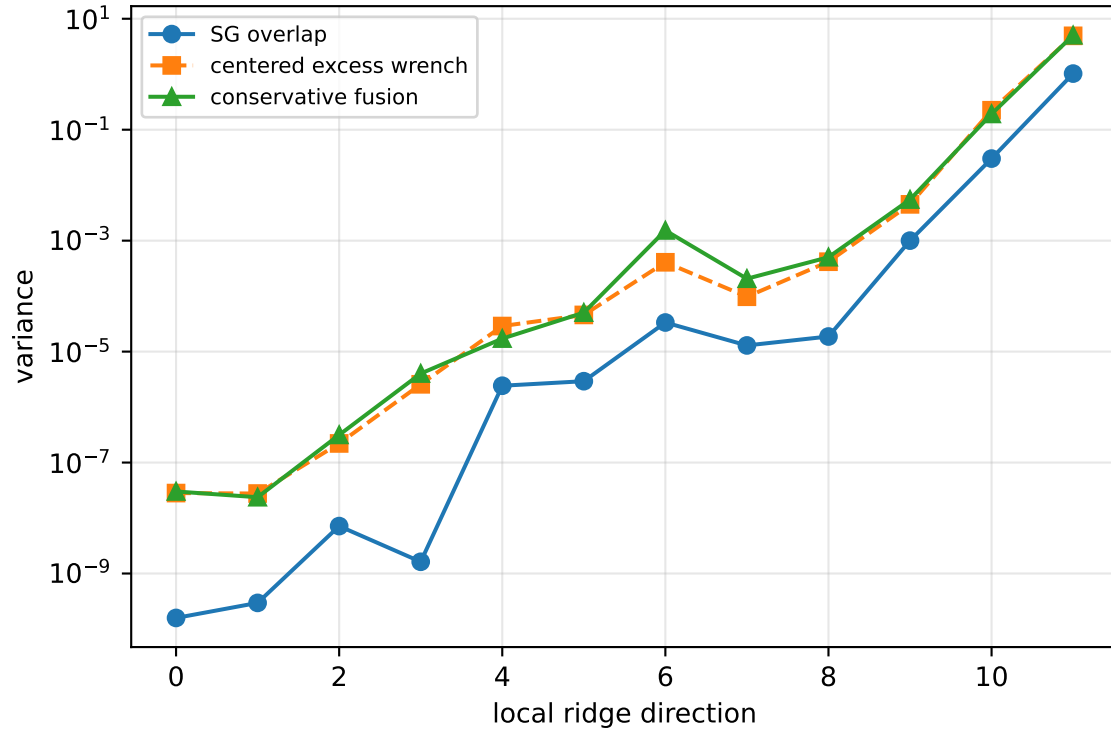
rotor lag cell: (0.221822739, 0.221826315] s

representative: 0.221824527 s

cog_all_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.34e-13)



cog_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 123.6$
wrench-to-acceleration recovery max error = $3.22\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

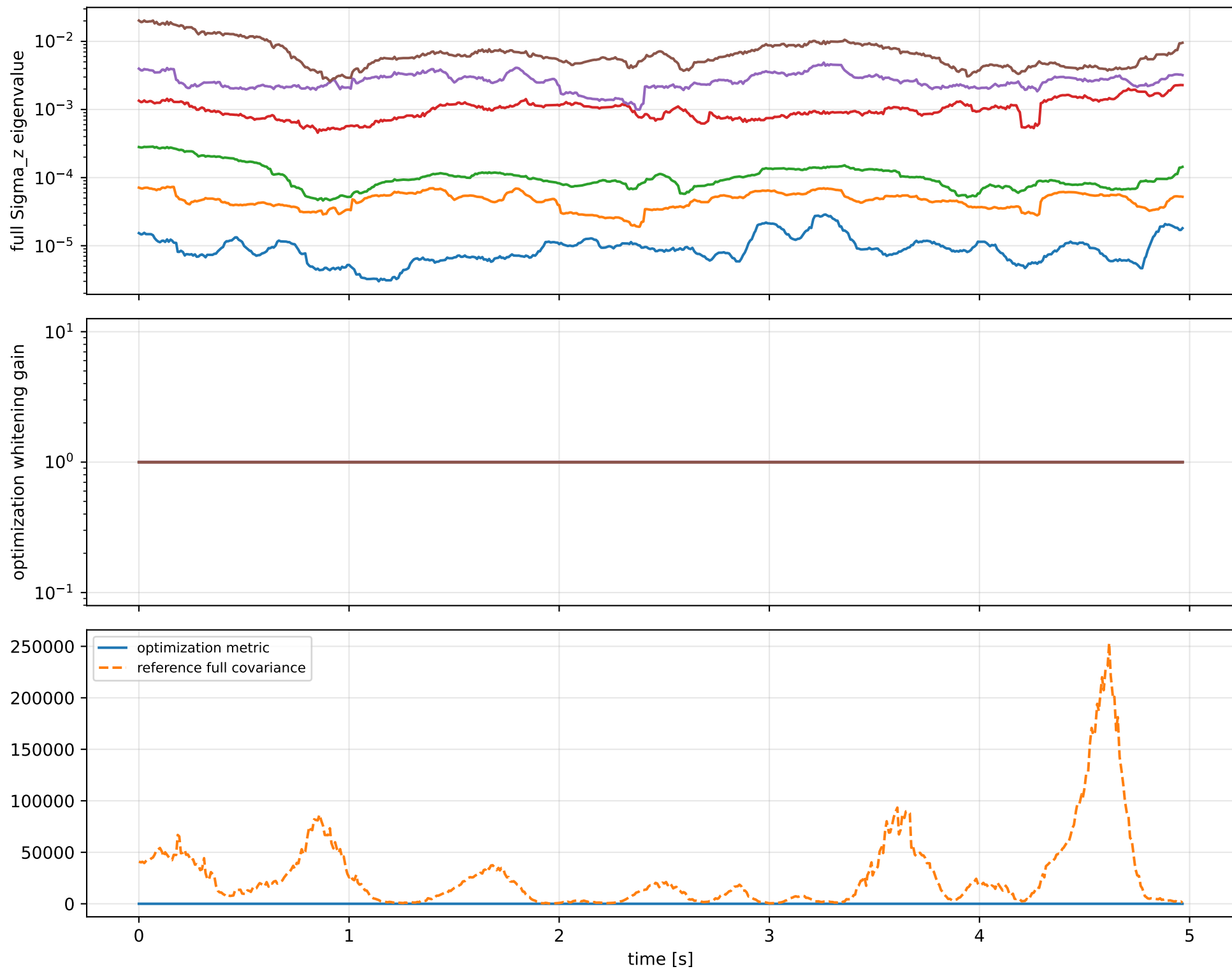
index 11: variance=5.035, ratio=4.902
 $\log_{\text{mass}} = -0.0557$; $\text{second_moment_diag_1} = -0.353$
 $\text{second_moment_diag_2} = +0.00237$; $\text{second_moment_diag_3} = +0.629$
 $\text{second_moment_offdiag_12} = +0.252$; $\text{second_moment_offdiag_13} = -0.597$
 $\text{second_moment_offdiag_23} = -0.212$; $\text{cog_x} = +2.25\text{e-}05$
 $\text{cog_y} = -0.000136$; $\text{cog_z} = +1.53\text{e-}05$
 $\log_{\text{force_eff_1}} = -0.0553$; $\log_{\text{force_eff_2}} = -0.0548$
 $\log_{\text{force_eff_3}} = -0.0564$; $\log_{\text{force_eff_4}} = -0.0562$

index 10: variance=0.1922, ratio=6.373
 $\log_{\text{mass}} = +0.0178$; $\text{second_moment_diag_1} = -0.0369$
 $\text{second_moment_diag_2} = +0.183$; $\text{second_moment_diag_3} = -0.235$
 $\text{second_moment_offdiag_12} = +0.848$; $\text{second_moment_offdiag_13} = +0.251$
 $\text{second_moment_offdiag_23} = -0.355$; $\text{cog_x} = +0.00173$
 $\text{cog_y} = -0.00359$; $\text{cog_z} = +0.000168$
 $\log_{\text{force_eff_1}} = +0.0249$; $\log_{\text{force_eff_2}} = +0.0385$
 $\log_{\text{force_eff_3}} = +0.0117$; $\log_{\text{force_eff_4}} = -0.00388$

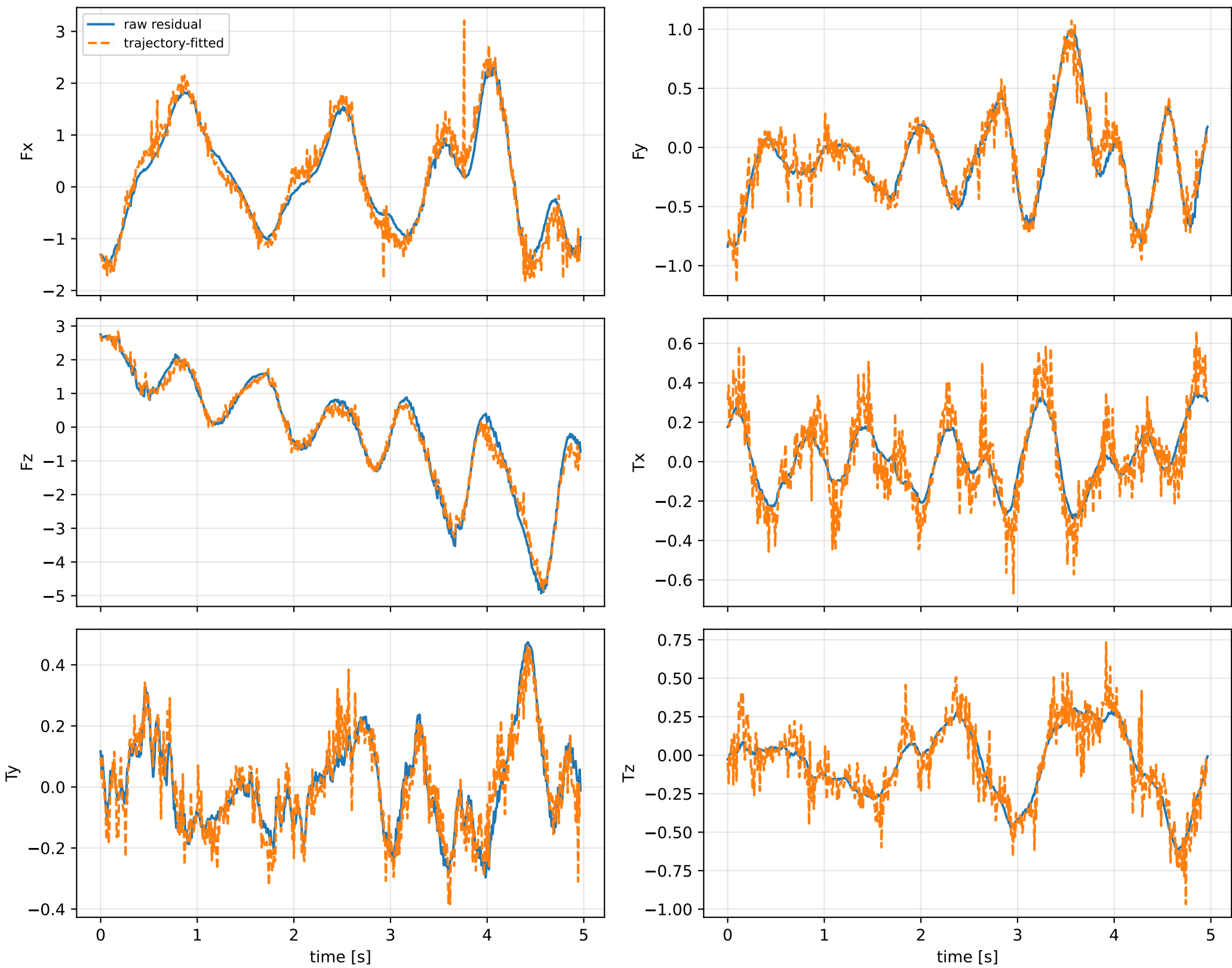
index 9: variance=0.005502, ratio=5.494
 $\log_{\text{mass}} = +0.159$; $\text{second_moment_diag_1} = +0.0226$
 $\text{second_moment_diag_2} = -0.9$; $\text{second_moment_diag_3} = +0.0825$
 $\text{second_moment_offdiag_12} = +0.0941$; $\text{second_moment_offdiag_13} = +0.102$
 $\text{second_moment_offdiag_23} = -0.186$; $\text{cog_x} = +0.00286$
 $\text{cog_y} = +0.00564$; $\text{cog_z} = +0.000225$
 $\log_{\text{force_eff_1}} = +0.12$; $\log_{\text{force_eff_2}} = +0.155$
 $\log_{\text{force_eff_3}} = +0.189$; $\log_{\text{force_eff_4}} = +0.172$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog_all_nominal: optimization vs reference covariance



cog_all_nominal: wrench / closure (force max 1.22e-14, torque max 5e-16)



cog_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_cog_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/cog_all_nominal.

SHA256: c8e09152b70075c9dea19c9858059b27978f15fa2ccc061d3e475ff0b4b7566a

data objective: 469.181561358

prior objective: 0.00748099247212

total objective: 469.189042351

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog_all_nominal (cog_position_body_m)

components: x, y, z

target: [-2.02470856e-03 -3.05265789e-05 9.50974960e-03]

std: [1.e-05 1.e-05 1.e-05]

final: [-2.02460688e-03 -3.04869956e-05 9.50853128e-03]

error: [1.01679240e-07 3.95832938e-08 -1.21831564e-06]

standardized residual: [0.01016792 0.00395833 -0.12183156]

factor objective: 0.00748099247212